

Activity 2: Applying the Object Oriented Systems Development life cycle (OODSLC).

Developing a Smart Library Management System

1) Requirement Analysis:

→ Objective: • Understand the needs of the college library.

• Gather requirements from librarians, students and administrators.

→ Activities: • Study the existing library system and collect user requirements.

• Identify functional requirements and non functional requirements.

→ Deliverables: • Software Requirement Specifications (SRS)

• Feasibility Report.

• Requirement document

2) Object Oriented Analysis:

- Objectives:
 - Model real world entities within the college library domain.
 - Understand relationships and business logic without technical constraints
- Activities: Identify objects, classes, attributes and relationships in the system.
- Deliverables: Analysis model, class list, object relationship document

3) Object - Oriented Design (OOD)

- Objective: Design the structure and architecture of the system.
- Activities: Design database and user interface and define class relationship and methods. To apply concepts like inheritance and encapsulation
- Deliverables: Design document, Database design, UI design, Detailed class diagrams.

4) Implementation

- Objective: Convert the design into a working software application.
- Activities: Write source code, Create database, Integrate system modules.
- Deliverables: Source code, executable application Database.

5) Deployment

- Objective: Install the software in the college library.
- Activities: Install the system, configure database, Train users.
- Deliverables: Installed software, User Manual, Training Guide.

6) Expected Learning Outcomes

- * Understand every phase of the OOSDLC.
- * Learn how object oriented concepts are used in software deployment.
- * Understand the importance of UML diagrams in analysis, design, implementation, testing and maintenance.